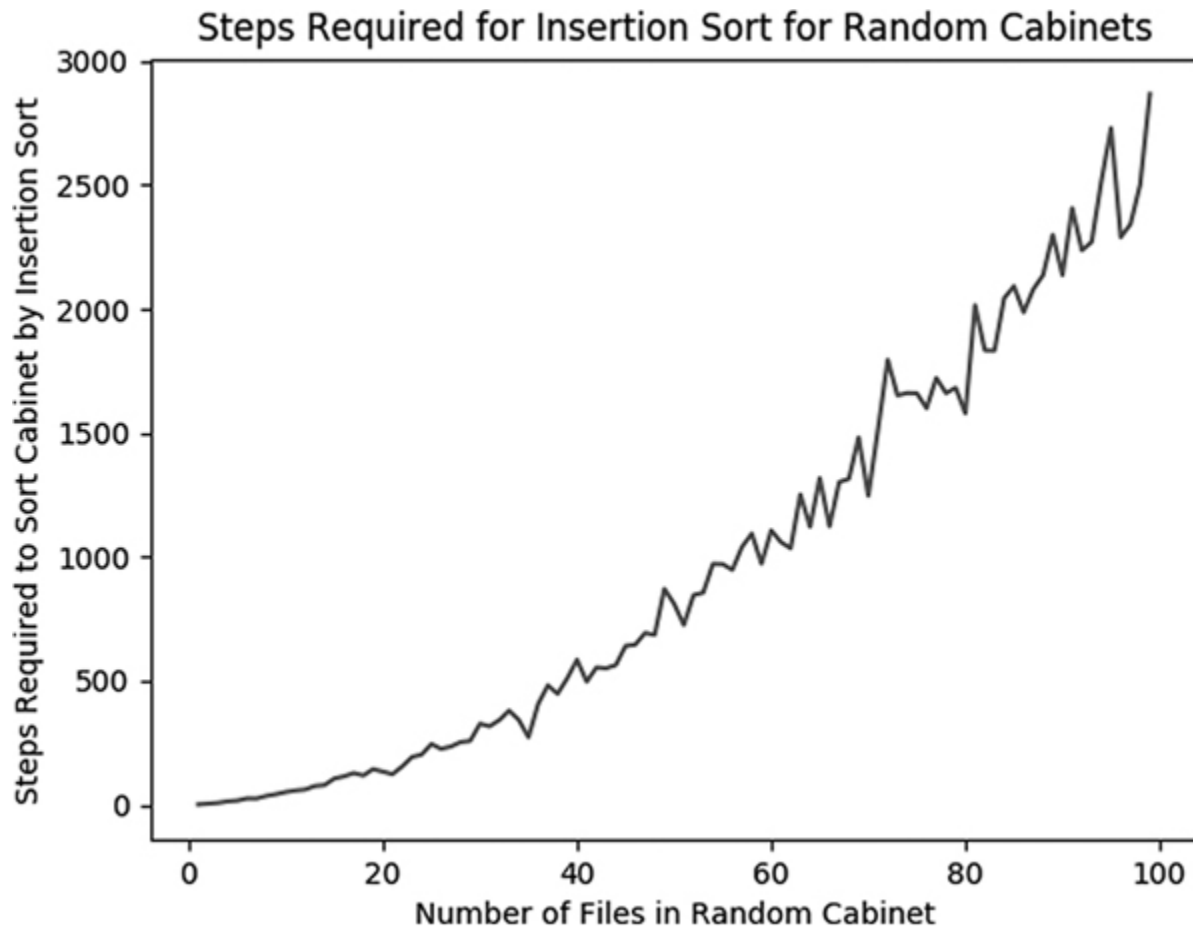


and occasionally it will create a list that is harder to deal with quickly, strictly through random chance. For this same reason, you may find that the output on your screen doesn't look exactly like the output printed here if you don't use the same random seed, but the general shape should be the same.



[Figure 4-1:](#) Insertion sort steps

Comparing to Well-Known Functions

Looking beyond the superficial jaggedness of [Figure 4-1](#), we can examine the general shape of the curve and try to reason about its growth rate. The number of steps required appears to grow quite slowly between $x = 1$ and about $x = 10$. After that, it seems to slowly get steeper (and more jagged). Between about $x = 90$ and $x = 100$, the growth rate appears very steep indeed.